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Past Papers by Waqar Siddhu
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When each outcome of a sample space has equal chance to occur as any other, the outcomes are called:

Answer (Please select your correct option)

VuAnswers.com

☐

Mutually exclusive

☒

Equally likely

☐

Not mutually exclusive

☐

Exhaustive

Made by: Waqar Siddhu

When each outcome of a sample space has equal chance to occur as any other, the outcomes are called:

Answer (Please select your correct option)



Mutually exclusive

VuAnswers.com



Equally likely



Not mutually exclusive



Exhaustive

Made by: Waqar Siddhu

An expected value of a random variable is equal to:

Answer (Please select your correct option)

☐ Variance

☒ Mean

☐ Standard deviation

☐ Covariance

VuAnswers.com

Made by: Waqar Siddhu

When $f(x)$ is continuous probability function, then $P(X = 1)$ is:

Answer (Please select your correct option)

☐ 1

☐ ∞

☐ $-\infty$

☒ 0

VuAnswers.com

Made by: Waqar Siddhu

The sum of deviations is zero, when deviations are taken from:

Answer (Please select your correct option)

☒ Mean

☐ Median

☐ Mode

☐ H.M

VuAnswers.com

Made by: Waqar Siddhu

The distribution function $F(x)$ is equal to

Answer (Please select your correct option)

VuAnswers.com

☐ $P(X = x)$

☒ $P(X \leq x)$

☐ $P(X \geq x)$

☐ $P(X > x)$

Made by: Waqar Siddhu

In a one-way ANOVA:

Answer (Please select your correct option)



The interaction term has $(c - 1)(n - 1)$ degrees of freedom



An interaction term is given



An interaction effect can be tested



There is no interaction term

VuAnswers.com

Made by: Waqar Siddhu

The degrees of freedom for a t-test with sample size 'n' is:

Answer (Please select your correct option)

☒ n-1

☐ n+1

☐ n-2

☐ n+2

VuAnswers.com

Made by: Waqar Siddhu

Rumour has reached the Trading Standards Officer that the manufacturer ABC is deliberately underfilling his cartons of orange juice. It is decided that a sample should be taken to check this claim. The stated contents on the carton are 100 ml on the average, then the null hypothesis is:

Answer (Please select your correct option)

☒ $H_0: \mu = 100$

☐ $H_0: \mu > 100$

☐ $H_0: \mu < 100$

☐ ~~$H_0: \mu \neq 100$~~

VuAnswers.com

Made by: Waqar Siddhu

When c is a constant, then $E(c)$ is:

Answer (Please select your correct option)

☐

0

☐

1

☒

c

☐

$-c$

VuAnswers.com

Made by: Waqar Siddhu

The combined distribution of more than two random variables is:

VuAnswers.com

Answer (Please select your correct option)

☐ Univariate distribution

☒ Joint distribution

☐ Marginal distribution

☐ Bivariate distribution

Made by: Waqar Siddhu

The test statistic used in analysis of variance procedure follow the :

Answer (Please select your correct option)

☐ χ^2 -distribution.

☐ T-distribution.

☐ Z-distribution.

☒ F-distribution.

VuAnswers.com

Made by: Waqar Siddhu

In normal distribution $\beta_2 = \dots\dots\dots$:

Answer (Please select your correct option)

VuAnswers.com

1



2



3



0



Made by: Waqar Siddhu

In normal distribution, the quartile deviation Q.D =

Answer (Please select your correct option)

☐ 0.5σ

☐ 0.75σ

☐ 0.7979σ

☒ 0.6745σ

VuAnswers.com

Made by: Waqar Siddhu

In a symmetrical distribution, the coefficient of skewness is equal to :

Answer (Please select your correct option)

☐ -1

☐ +1

☒ 0

☐ 2

VuAnswers.com

Made by: Waqar Siddhu

The average which is defined as the reciprocal of the arithmetic mean of the reciprocals of the values is called:

Answer (Please select your correct option)

☐ Geometric Mean

☒ Harmonic Mean

☐ Mode

☐ Median

VuAnswers.com

Made by: Waqar Siddhu

Which measure of dispersion is used to compare the variation of two data sets?

Answer (Please select your correct option)

☒ Coefficient of variation

☐ Coefficient of comparison

☐ Mean deviation

☐ Standard deviation

VuAnswers.com

Made by: Waqar Siddhu

If $S.D(X) = 5$ then $S.D(\frac{2X+5}{2}) = \underline{\hspace{2cm}}$

Answer (Please select your correct option)

☐ 5

☐ 10

☐ 15

☒ 7.5

VuAnswers.com

Made by: Waqar Siddhu

The deviation of a distribution from symmetry is called:

Answer (Please select your correct option)

☐ Kurtosis

☒ Skewness

☐ Dispersion

☐ Flatness

VuAnswers.com

Made by: Waqar Siddhu

The conditional probability function $f(x|1) =$ _____

Answer (Please select your correct option)

VuAnswers.com

☐ $f(1,1)$

☐ $f(x,1)$

☒ $\frac{f(x,1)}{h(1)}$

☐ $\frac{f(x,1)}{h(x)}$

Made by: Waqar Siddhu

Which one is the correct formula to find the desired sample size?

Answer (Please select your correct option)

VuAnswers.com

☒
$$n = \left(\frac{Z_{\alpha/2} \cdot \sigma}{e} \right)^2$$

☐
$$n = \left(\frac{Z_{\alpha/2} \cdot \sqrt{\sigma}}{e} \right)^2$$

X

☐
$$n = \left(\frac{Z_{\alpha/2} \cdot \bar{X}}{e} \right)^2$$

☐
$$n = \frac{Z_{\alpha/2} \cdot \sigma}{e}$$

Made by: Waqar Siddhu

A deserving player is not selected in the team is an example of

Answer (Please select your correct option)

☐ Type I error

☒ Type II error

☐ Correct decision

☐ No information regarding this

VuAnswers.com

Made by: Waqar Siddhu

A judge can acquit a guilty person is the example of

Answer (Please select your correct option)

☐ Type I error

☐ Type II error

☒ Correct decision

not sure

☐ No information regarding this

VuAnswers.com

Made by: Waqar Siddhu

Ideally, the width of confidence interval should be:

Answer (Please select your correct option)

☒ 0

☐ 1

☐ 99

☐ 100

VuAnswers.com

Made by: Waqar Siddhu

If the sampling distribution of $\bar{X}$ is normal, we would expect 99% of the sample means to be within the interval:

Answer (Please select your correct option)

VuAnswers.com

☐ $\mu_x \pm 2\sigma_x$

☐ $\mu_x \pm 1.96\sigma_x$

☒ $\mu_x \pm 2.58\sigma_x$

☐ $\mu_x \pm \sigma_x$

Made by: Waqar Siddhu

If mean of χ^2 distribution is k then variance will be:

Answer (Please select your correct option)

VuAnswers.com

☐ k^2

☒ $2k$

☐ $1/k$

☐ k

Made by: Waqar Siddhu

Mean of the F-distribution is possible only, when

Answer (Please select your correct option)

VuAnswers.com

☐ $v_1 > 2$

☒ $v_2 > 2$

☐ $v_1 < 2$

☐ $v_2 < 2$

Made by: Waqar Siddhu

In Statistics, we have MSE which is abbreviation of

Answer (Please select your correct option)

☒ Mean square error

☐ Measured square error

☐ Medical screening exam

☐ Major sampling error

VuAnswers.com

Made by: Waqar Siddhu

What is the graphical shape of the chi-square distribution?

Answer (~~Please select your correct option~~)

☒ Positively skewed

☐ Negatively skewed

☐ Uniformly distributed

☐ Normally distributed

VuAnswers.com

Made by: Waqar Siddhu

As the degree of freedom increases, the t-distribution tends to coincide with:

Answer (Please select your correct option)

☐ Binomial distribution

☐ Uniform distribution

☐ Hypergeometric distribution

☒ Normal distribution

VuAnswers.com

Made by: Waqar Siddhu

If X and Y are independent variables, then $E(XY)$ is:

Answer (Please select your correct option)

☐ $E(X)$

☒ $E(X) \cdot E(Y)$

☐ $X \cdot E(Y)$

☐ $Y \cdot E(X)$

VuAnswers.com

Made by: Waqar Siddhu

What are the number of ways in which four books can be arranged on a shelf?

Answer (Please select your correct option)

☐ 4

☐ 6

☐ 12

☒ 24

VuAnswers.com

Made by: Waqar Siddhu

The parameters of the binomial distribution $b(x, n, p)$ are:

Answer (Please select your correct option)

☐ x & n

☐ x & p

☒ n & p

☐ x, n & p

VuAnswers.com

Made by: Waqar Siddhu

Hypergeometric probability distribution has :

Answer (Please select your correct option)

☐ (n, k) parameter

☐ (N) parameter

☐ (N, n, N-k) parameter

☒ (N, n, k) parameter

VuAnswers.com

Made by: Waqar Siddhu

The variance of the hypergeometric probability distribution is:

Answer (Please select your correct option)

☐ $\sigma^2 = n \frac{k}{N}$

☒ $\sigma^2 = n \frac{k}{N} \frac{N-k}{N} \frac{N-n}{N-1}$

☐ $\sigma^2 = npq$

☐ $\sigma^2 = N \frac{n}{K}$

VuAnswers.com

Made by: Waqar Siddhu

When $f(x)$ is continuous probability function for $1 < X < 5$, then $P(X < 1)$ is:

Answer (Please select your correct option)

☐

0

☐

0.25

☐

0.5

☐

1

VuAnswers.com

Made by: Waqar Siddhu

For any two estimators T_1 and T_2 , if $VAR(T_1) < VAR(T_2)$, then T_1 is:

Answer (Please select your correct option)

☐ Unbaised

☐ Sufficient

☒ Efficient

☐ Consistent

VuAnswers.com

Made by: Waqar Siddhu

If an estimator gets closer to the population parameter by increasing sample size then it is known as:

Answer (Please select your correct option)

☒ Consistent estimator

☐ Sufficient estimator

☐ Efficient estimator

☐ Unbiased estimator

VuAnswers.com

Made by: Waqar Siddhu

Which of the following comes first to make frequency distribution.

Answer (Please select your correct option)

☐ Number of Groups

☐ Class interval

☒ Range

☐ Tally marks

VuAnswers.com

Made by: Waqar Siddhu

What curve shape would you expect for the distribution of death rates of population of all age groups?

Answer (Please select your correct option)

☐ Symmetrical curve

☐ Skewed to the right

☐ Skewed to the left

☒ U shape curve

VuAnswers.com

Made by: Waqar Siddhu

Which one is the measure of central tendency:

Answer (Please select your correct option)

☐ Variation of the distribution

☒ Average of the distribution

☐ Scatterness of the distribution

☐ Dispersion of the distribution

VuAnswers.com

Made by: Waqar Siddhu

The parameter of the chi- square distribution is.....

Answer (Please select your correct option)

VuAnswers.com



ν



$\nu - 1$



$\nu - 2$



$\nu - p$

Made by: Waqar Siddhu

The mean of the F-distribution is:

Answer (Please select your correct option)

VuAnswers.com

☐ $\frac{v_1}{v_1 - 2}$ for $v_1 > 2$

☒ $\frac{v_2}{v_2 - 2}$ for $v_2 > 2$

☐ $\frac{v_1}{v_1 - 2}$ for $v_1 \geq 2$

☐ $\frac{v_2}{v_2 - 2}$ for $v_1 \leq 2$

Made by: Waqar Siddhu

The F-distribution always ranges from:

Answer (Please select your correct option)

VuAnswers.com

☐ 0 to 1

☐ 0 to $-\infty$

☐ $-\infty$ to $+\infty$

☒ 0 to $+\infty$

Made by: Waqar Siddhu

An expected value of a random variable is equal to:

Answer (Please select your correct option)

VuAnswers.com

☐ Variance

☒ Mean

☐ Standard deviation

☐ Covariance

Made by: Waqar Siddhu

When $f(x)$ is continuous probability function, then $P(X = 1)$ is:

Answer (Please select your correct option)

VuAnswers.com

☐

1

☐

∞

☐

$-\infty$

☒

0

Made by: Waqar Siddhu

Rumour has reached the Trading Standards Officer that the manufacturer ABC is deliberately underfilling his cartons of orange juice. It is decided that a sample should be taken to check this claim. The stated contents on the carton are 100 ml on the average, then the alternative hypothesis is:

Answer (Please select your correct option)

VuAnswers.com

☐ $H_1: \mu = 100$

☐ $H_1: \mu > 100$

☐ $H_1: \mu < 100$

☒ $H_1: \mu \neq 100$

Made by: Waqar Siddhu



Which of the following is a characteristics of the normal distribution:

Answer (Please select your correct option)

VuAnswers.com

☐ It is a skewed distribution

☒ It is bell-shaped

☐ It is not asymptotic

☐ It is leptokurtic

Made by: Waqar Siddhu

For the given poisson distribution $P(X = 1) = \frac{e^{-0.135} 0.135^1}{1!}$ the mean value is :

Answer (Please select your correct option)

VuAnswers.com

☐ $e^{-0.135}$

☐ -0.135

☒ 0.135

☐ 1

Made by: Waqar Siddhu

In normal distribution $\beta_2 = \dots\dots\dots$:

Answer (Please select your correct option)

VuAnswers.com

☐ 1

☐ 2

☒ 3

☐ 0

Made by: Waqar Siddhu

In normal distribution, the quartile deviation Q.D =

Answer (Please select your correct option)

VuAnswers.com

- ☐ 0.5σ
- ☐ 0.75σ
- ☐ 0.7979σ
- ☒ 0.6745σ

Made by: Waqar Siddhu

A good way to get a small standard error is to use a _____.

Answer (Please select your correct option)

**logical quiez not
sure**

VuAnswers.com

☐ Repeated sampling

☐ Small sample

☒ Large sample

☐ Large population

mind it not sure

Made by: Waqar Siddhu

The difference between the largest and the smallest data values is called the

$$x_m - x_0$$

Answer (Please select your correct option)

VuAnswers.com

☐ Variance

$$x_m = \max \text{ rane } \quad x_0 =$$

☐ Interquartile range

$$\min \text{ range}$$

☒ Range

☐ Coefficient of variation

Made by: Waqar Siddhu

Which is appropriate average for finding the average speed of a car:

Answer (Please select your correct option)

VuAnswers.com

☐ Mean

☐ Geometric mean

☒ Harmonic mean

☐ Weighted mean

Made by: Waqar Siddhu

Which one is the formula of mid range:

Answer (Please select your correct option)

VuAnswers.com

☐ $x_m - x_0$

☐ $x_0 - x_m$

☐ $\frac{x_0 - x_m}{2}$

☒ $\frac{x_0 + x_m}{2}$

Made by: Waqar Siddhu

Which one of the following is a meso-kurtic curve?

Answer (Please select your correct option)

VuAnswers.com

☐ Negatively skewed

☐ Positively skewed

☐ J-shaped

☒ Normal

Made by: Waqar Siddhu



If you draw all possible samples from some population, calculate the mean for each of the sample and construct the probability distribution of the sample means, what would you have?

Answer (Please select your correct option)

VuAnswers.com

☐ A population distribution

Logical not sure

☐ A sample distribution

☒ A sampling distribution

☐ A parameter distribution

Made by: Waqar Siddhu

By definition $f(y | x) =$ _____

Answer (Please select your correct option)

VuAnswers.com

☐ $f(y)$

☐ $f(x, y)$

☒ $\frac{f(x, y)}{h(x)}$

☐ $\frac{f(x, y)}{h(y)}$

Made by: Waqar Siddhu

The critical region for $H_1 : \mu > \mu_0$ when $\alpha = 0.01$ is:

not sure

Answer (Please select your correct option)

VuAnswers.com

☒ $z > z_{0.01}$

☐ $|z| > z_{0.01}$

☐ $z < -z_{0.05}$

☐ $|z| > z_{0.05}$

Made by: Waqar Siddhu

For degree of freedom $\nu > 2$ the variance of t-distribution is always:

Answer (Please select your correct option)

VuAnswers.com

☐ Greater than zero

☐ Less than one

☐ Equal to one

☒ Greater than one

Made by: Waqar Siddhu

Ideally, the width of confidence interval should be:

Answer (Please select your correct option)

VuAnswers.com

☒ 0

☐ 1

☐ 99

☐ 100

Made by: Waqar Siddhu

If the sampling distribution of $\bar{X}$ is normal, we would expect 99% of the sample means to be within the interval:

Answer (Please select your correct option)

VuAnswers.com

☐ $\mu_x \pm 2\sigma_x$

☐ $\mu_x \pm 1.96\sigma_x$

☒ $\mu_x \pm 2.58\sigma_x$

☐ $\mu_x \pm \sigma_x$

Made by: Waqar Siddhu

Which one of the formula will be used to find out the confidence interval for μ , when population variance unknown and sample size is large?

Answer (Please select your correct option)

VuAnswers.com

☒ $\bar{x} \pm Z_{\alpha/2} \frac{s}{\sqrt{n}}$

☐ $\bar{x} \pm t_{\alpha/2(v)} \frac{s}{\sqrt{n}}$

☐ $\bar{x} \pm t_{\alpha/2(v)} \frac{\sigma}{\sqrt{n}}$

☐ $\bar{x} \pm Z_{\alpha/2} \frac{s}{\sqrt{n-1}}$

Made by: Waqar Siddhu

If $\bar{X}$ is the mean of the n observations, then which test statistic will be used to calculate the confidence limits of the population variance σ^2 ?

Answer (Please select your correct option)

VuAnswers.com

☒ Z-statistic

☐ T-statistic

☐ χ^2 -statistics

☐ F-statistics

Made by: Waqar Siddhu

In Statistics, we have MSE which is abbreviation of

Answer (Please select your correct option)

VuAnswers.com

☒ Mean square error

☐ Measured square error

☐ Medical screening exam

☐ Major sampling error

Made by: Waqar Siddhu

In the test of goodness of fit, the _____ is used as a test statistic.

Answer (Please select your correct option)

VuAnswers.com

☐ F

☐ t

☐ Z

☒ χ^2

Made by: Waqar Siddhu

If there are 8 treatments with 6 blocks in a randomized completed block design then what are the degrees of freedom for treatments?

$$v=n-1$$

v represents the

degree of freedom

VuAnswers.com

Answer (Please select your correct option)

5



4



6



7



Made by: Waqar Siddhu

What factor determines the shape of the t-distribution?

Answer (Please select your correct option)

VuAnswers.com

☒ Degree of freedom

☐ Critical value

☐ Frequency of data

☐ Probability

Made by: Waqar Siddhu

If X and Y are random variables, then $E(X - Y)$ is equal to:

Answer (Please select your correct option)

VuAnswers.com

☐ $E(X) + E(Y)$

☒ $E(X) - E(Y)$

☐ $X - E(Y)$

☐ $E(X) - Y$

Made by: Waqar Siddhu

Two continuous r.v.'s X and Y are said to be independent if and only if :

Answer (Please select your correct option)

VuAnswers.com

☒ $f(x,y) = g(x) h(y)$

☐ $f(x,y) \neq g(x) h(y)$

☐ $f(x,y) > g(x) h(y)$

☐ $f(x,y) < g(x) h(y)$

Made by: Waqar Siddhu



The lottery tickets issued for the purpose of money-making follows a:

Answer (Please select your correct option)

VuAnswers.com

☐ Normal distribution

☒ Discrete uniform distribution

☐ Binomial distribution

☐ Hypergeometric distribution

Made by: Waqar Siddhu

Uniform distribution is defined by:

Answer (Please select your correct option)

VuAnswers.com

☐ Largest value

☐ Largest and smallest value

☐ Smallest value

☒ Central value

Made by: Waqar Siddhu

If an estimator gets closer to the population parameter by increasing sample size then it is known as:

Answer (Please select your correct option)

VuAnswers.com

☒ Consistent estimator

☐ Sufficient estimator

☐ Efficient estimator

☐ Unbiased estimator

Made by: Waqar Siddhu

Quantitative variable is further divided into:

Answer (Please select your correct option)

VuAnswers.com

☐ Continuous variable

☐ Discrete variable

☒ Continuous & Discrete variable

☐ None of the above

Made by: Waqar Siddhu

Color of the dress is the example of

Answer (Please select your correct option)

VuAnswers.com

☒ Qualitative data

☐ Quantitative data

☐ Continuous data

☐ Discrete data

Made by: Waqar Siddhu

Which one is commonly called a bell shaped distribution?

Answer (Please select your correct option)

VuAnswers.com

☒ Symmetrical

☐ Bimodal

☐ Skewed

☐ U shaped

Made by: Waqar Siddhu

A fair coin is tossed three times, the probability that at least one head appear is:

Answer (Please select your correct option)

VuAnswers.com

☐ $1/2$

☐ $1/8$

☐ $6/8$

☒ $7/8$

Made by: Waqar Siddhu

The probability of simultaneous occurrence of two events is called:

Answer (Please select your correct option)

VuAnswers.com

☐ Subjective probability

☐ Conditional probability

☒ Joint probability

☐ Prior probability

Made by: Waqar Siddhu

What is the stem part of 243:

Answer (Please select your correct option)

VuAnswers.com

3



43



23



24



Made by: Waqar Siddhu

A numerical value used as a summary measure for a sample, such as sample mean, is known as a :

Answer (Please select your correct option)

VuAnswers.com

☐ Population Parameter

☐ Sample Parameter

☐ Sample Statistic

☒ Population mean

Made by: Waqar Siddhu

Given the series 1,2,1,1,2,2,2,2,3,4,5,3,2,3,1,4,2,3. Which one of the following is mode of the given series:

Answer (Please select your correct option)

VuAnswers.com

4

☐

3

☐

3

2

☐

3

2

1

☒

3

2*

repeted values is called

Made by: Waqar Siddhu



$P(A \text{ or } B) = P(A) + P(B)$, then A and B are:

Answer (Please select your correct option)

VuAnswers.com

☒ Mutually exclusive events

☐ Independent events

☐ Exhaustive events

☐ Equally likely events

Made by: Waqar Siddhu



First moment about origin is always equals to:

Answer (Please select your correct option)

VuAnswers.com

☐

Mean

☐

Variance

☐

Standard Deviation

☒

Zero

Made by: Waqar Siddhu

$$E(4X + 5) = \underline{\hspace{2cm}}$$

Answer (Please select your correct option)

VuAnswers.com

☐ $16 E(X)$

☒ $16 E(X) + 5$

☐ $12 E(X)$

☐ $4 E(X) + 5$

Made by: Waqar Siddhu



When two coins are tossed the probability of at least one head is:

Answer (Please select your correct option)

VuAnswers.com



1/4



3/4



2/4



4/4

Made by: Waqar Siddhu

If σ^2 is unknown, then we use Z-test if the sample size is:

Answer (Please select your correct option)

VuAnswers.com

☒ $n \geq 30$ 

☐ $n < 30$

☐ $n = 25$

☐ $n = 20$

Made by: Waqar Siddhu



When a coin is tossed 3 times, the probability of 3 tails is:

Answer (Please select your correct option)

VuAnswers.com

☒ 1/8

☐ 2/4

☐ 3/8

☐ 2/8

Made by: Waqar Siddhu



The F-distribution has parameter.

Answer (Please select your correct option)

VuAnswers.com

One



No



Two



Three



Made by: Waqar Siddhu



Which one of the following provides the basis for hypothesis testing?

Answer (Please select your correct option)

VuAnswers.com



Null hypothesis



Alternative hypothesis



Critical value



Test-statistic

Made by: Waqar Siddhu



Rumour has reached the Trading Standards Officer that the manufacturer ABC is deliberately underfilling his cartons of orange juice. It is decided that a sample should be taken to check this claim. The stated contents on the carton are 100 ml on the average, then the alternative hypothesis is:

Answer (Please select your correct option)

VuAnswers.com

☐ $H_1: \mu = 100$

☐ $H_1: \mu > 100$

☐ $H_1: \mu < 100$

☒ $H_1: \mu \neq 100$

Made by: Waqar Siddhu

Rumour has reached the Trading Standards Officer that the manufacturer ABC is deliberately underfilling his cartons of orange juice. It is decided that a sample should be taken to check this claim. The stated contents on the carton are 100 ml on the average, then the null hypothesis is:

Answer (Please select your correct option)

VuAnswers.com

☒ $H_0 : \mu = 100$

☐ $H_0 : \mu > 100$

☐ $H_0 : \mu < 100$

☐ $H_0 : \mu \neq 100$

Made by: Waqar Siddhu

By definition $f(x_i | y_j) =$ _____

VuAnswers.com

Answer (Please select your correct option)

☒ $\frac{f(x_i, y_j)}{h(y_j)}$

☐ $\frac{f(x_i, y_j)}{h(x_i)}$

☐ $f(x_i, y_j)$

☐ $f(y_j)$

Made by: Waqar Siddhu



The test statistic used in analysis of variance procedure follow the :

Answer (Please select your correct option)

VuAnswers.com



χ^2 -distribution.



T-distribution.



Z-distribution.



F-distribution.

Made by: Waqar Siddhu



Which one of the following is the most common example of a situation for which the main parameter of interest is a population proportion?

Answer (Please select your correct option)

VuAnswers.com



An observational study



A normal experiment



A binomial experiment

Made by: Waqar Siddhu



An estimator which has the smallest standard error among all unbiased estimators fulfills the property of _____.

Answer (Please select your correct option)

VuAnswers.com

☐ Unbiasedness

☒ Efficiency

☐ Consistency

☐ Sufficiency

Made by: Waqar Siddhu



An estimator which has the smallest standard error among all unbiased estimators fulfills the property of _____.

Answer (Please select your correct option)

VuAnswers.com



Unbiasedness



Efficiency



Consistency



Sufficiency

Made by: Waqar Siddhu



Which of the following can never be taken as the probability of an event?

Answer (Please select your correct option)

VuAnswers.com



1



0



0.5



-0.5

Made by: Waqar Siddhu



A set is any well-defined collection of

Answer (Please select your correct option)

VuAnswers.com



Positive Objects



Negative Objects



Same Objects



Distinct Objects

Made by: Waqar Siddhu

Measure of dispersion is used to calculate the:

Answer (Please select your correct option)

VuAnswers.com

☐ Central value

☐ Highest value

☐ Lowest value

☒ Scattered value

Made by: Waqar Siddhu



If X and Y are independent variables then $\text{Var}(X-Y) =$ _____

Answer (Please select your correct option)

VuAnswers.com

☒ $\text{Var}(X) - \text{Var}(Y)$

☐ $\text{Var}(X) + \text{Var}(Y)$

☐ $\text{Var}(X+Y)$

☐ $\text{Var}(X) \times \text{Var}(Y)$



Made by: Waqar Siddhu

How the standard error is decreased :

Answer (Please select your correct option)

VuAnswers.com

☐

By decreasing the sample size

☐

By decreasing the mean

☐

By increasing the standard deviation

☒

By increasing the sample size

Made by: Waqar Siddhu



The total number of samples when sampling is done with replacement is equal to:

Answer (Please select your correct option)

VuAnswers.com



$$N^n$$



$$C_n^N$$



$$\frac{N-n}{N-1}$$



$$1$$

Made by: Waqar Siddhu



If you draw all possible samples from some population, calculate the mean for each of the sample and construct the probability distribution of the sample means, what would you have?

Answer (Please select your correct option)

VuAnswers.com



A population distribution



A sample distribution



A sampling distribution



A parameter distribution

Made by: Waqar Siddhu



The conditional probability function $f(x|1) =$ _____

Answer (Please select your correct option)

VuAnswers.com

☐

$$f(1,1)$$

☐

$$f(x,1)$$

☒

$$\frac{f(x,1)}{h(1)}$$

☐

$$\frac{f(x,1)}{h(x)}$$

Made by: Waqar Siddhu

For $\alpha = 0.01$, the critical values of z for two tailed test are equal to:

Answer (Please select your correct option)

VuAnswers.com

☒ -2.58 and +2.58



☐ -2.33 and +2.33

☐ -1.645 and +1.645

☐ -1.96 and + 1.96

Made by: Waqar Siddhu

A deserving player is not selected in the team is an example of:

Answer (Please select your correct option)

VuAnswers.com

☒ Type I error

☐ Type II error

☐ Correct decision

☐ No information regarding this

Made by: Waqar Siddhu



"A point estimate plus/minus a few times the standard error of that estimate". This statement represents:

Answer (Please select your correct option)

VuAnswers.com

☒ Confidence interval

☐ Critical region

☐ Acceptance region

☐ Critical value

Made by: Waqar Siddhu



The proportion of males in Pakistan is at least 0.48, the alternative hypothesis H_1 is

Answer (Please select your correct option)

VuAnswers.com



$$P \leq 0.48$$



$$P = 0.48$$



$$P < 0.48$$



$$P \geq 0.48$$

Made by: Waqar Siddhu



If $\bar{X}$ is the mean of the n observations, then which test statistic will be used to calculate the confidence limits of the population variance σ^2 ?

Answer (Please select your correct option)

VuAnswers.com



Z-statistic



T-statistic



χ^2 -statistics

not sure



F-statistics

Made by: Waqar Siddhu



To find the confidence interval for the ratio of two variances, we use

Answer (Please select your correct option)

VuAnswers.com

☒ F-Distribution

☐ Z-Distribution

☐ Chi-square-Distribution

☐ t-Distribution

Made by: Waqar Siddhu



The Chi- Square distribution is continuous distribution ranging from:

Answer (Please select your correct option)

VuAnswers.com



$$-\infty \leq \chi^2 \leq \infty$$



$$-\infty \leq \chi^2 \leq 1$$



$$-\infty \leq \chi^2 \leq 0$$



$$0 \leq \chi^2 \leq \infty$$

Made by: Waqar Siddhu



In Statistics, we have MSE which is abbreviation of

Answer (Please select your correct option)

VuAnswers.com

☒ Mean square error

☐ Measured square error

☐ Medical screening exam

☐ Major sampling error

Made by: Waqar Siddhu



In a binomial experiment the total number of trials are:

Answer (Please select your correct option)

VuAnswers.com

☒ Fixed in advance

☐ Changeable according to situation

☐ Unpredictable

☐ Not independent

Made by: Waqar Siddhu



The lottery tickets issued for the purpose of money-making follows a:

Answer (Please select your correct option)

VuAnswers.com

☐ Normal distribution

☒ Discrete uniform distribution

☐ Binomial distribution

☐ Hypergeometric distribution

Made by: Waqar Siddhu



Which of the following value could not represent a coefficient of correlation?

Answer (Please select your correct option)

VuAnswers.com



$$r = 0.99$$



$$r = 1.09$$



$$r = -0.73$$



$$r = -1$$

Made by: Waqar Siddhu



In a one way ANOVA test there are 5 observations in each of three treatments. The degrees of freedom for the treatments is:

Answer (Please select your correct option)

VuAnswers.com

5



3



1



2



Made by: Waqar Siddhu



If $P(B|A) = 0.25$ and $P(A \cap B) = 0.20$, then $P(A) =$

Answer (Please select your correct option)

VuAnswers.com



0.05



0.80



0.95



0.75

Made by: Waqar Siddhu



If a random variable X denotes the number of heads when three distinct coins are tossed, the X assumed the values:

Answer (Please select your correct option)

VuAnswers.com

☒ 0,1,2,3

☐ 1,3,3,1

☐ 1, 2, 3

☐ 3, 2

Made by: Waqar Siddhu



When $f(x)$ is continuous probability function, then $P(X = 2)$ is:

Answer (Please select your correct option)

VuAnswers.com

☐

1

☐

0.5

☒

0

☐

0.25

Made by: Waqar Siddhu



Atmosphere pressure is the example of:

Answer (Please select your correct option)

VuAnswers.com



Constant



Qualitative variable



Quantitative variable



None of the above

Made by: Waqar Siddhu



Which of the following scale has true zero point?

Answer (Please select your correct option)

VuAnswers.com

☒ Ratio Scale

☐ Interval scale

☐ Nominal scale

☐ Ordinal scale

Made by: Waqar Siddhu



Given the series 1,2,1,1,2,2,2,2,3,4,5,3,2,3,1,4,2,3. Which one of the following is mode of the given seires:

Answer (Please select your correct option)

VuAnswers.com

☐

4

☐

3

3

☐

2

3

2

☒

1

3

2*

Made by: Waqar Siddhu

Var($4X + 5$) = _____

Answer (Please select your correct option)

VuAnswers.com

☐ 16 Var (X)

☐ 16 Var (X) + 5

☒ 4 Var (X) + 5

☐ 12 Var (X)

Made by: Waqar Siddhu

When $f(x)$ is continuous probability function, then $P(X = 1)$ is:

Answer (Please select your correct option)

VuAnswers.com

☐ 1

☐ ∞

☐ $-\infty$

☒ 0

Made by: Waqar Siddhu

The hyper geometric random variable is a(an):

Answer (Please select your correct option)

VuAnswers.com

☐ Continuous variable

☒ Discrete variable

☐ Undefined

☐ Independent variable

Made by: Waqar Siddhu

The degrees of freedom for a t-test with sample size 'n' is:

Answer (Please select your correct option)

☒ n-1

☐ n+1

☐ n-2

☐ n+2

VuAnswers.com

Made by: Waqar Siddhu

Rumour has reached the Trading Standards Officer that the manufacturer ABC is deliberately underfilling his cartons of orange juice. It is decided that a sample should be taken to check this claim. The stated contents on the carton are 100 ml on the average, then the null hypothesis is:

Answer (Please select your correct option)

VuAnswers.com

☒ $H_0: \mu = 100$

☐ $H_0: \mu > 100$

☐ $H_0: \mu < 100$

☐ $H_0: \mu \neq 100$

Made by: Waqar Siddhu

Assuming that following is a probability distribution, then what is the value of 'a':

X	1	2	3
P(X)	0.1	a	0.1

Answer (Please select your correct option)

VuAnswers.com

☐ 0.6

☒ 0.8

☐ 0.2

☐ 0.4

Made by: Waqar Siddhu

In normal distribution $\beta_1 = \dots\dots\dots$:

Answer (Please select your correct option)

VuAnswers.com

☒ 0

☐ 1

☐ 2

☐ 3

Made by: Waqar Siddhu

A good way to get a small standard error is to use a _____.

Answer (Please select your correct option)

VuAnswers.com

☐ Repeated sampling

☐ Small sample

☒ Large sample

☐ Large population

Made by: Waqar Siddhu

A randomly selected sample of 400 students at university was asked whether or not they will participate in politics. Forty-six percent of the 400 student surveyed answered "yes". Which one of the following statement about number 46% is correct?

Answer (Please select your correct option)

VuAnswers.com

☐ It is a sample statistic.

☐ It is a population parameter.

☐ It is a margin of error.

☒ It is a standard error.

not sure

Made by: Waqar Siddhu

If you draw all possible samples from some population, calculate the mean for each of the sample and construct the probability distribution of the sample means, what would you have?

Answer (Please select your correct option)

VuAnswers.com

☐ A population distribution

☒ A sample distribution

☒ A sampling distribution

☐ A parameter distribution

Made by: Waqar Siddhu

The conditional probability function $f(x|1) =$ _____

Answer (Please select your correct option)

VuAnswers.com

☐ $f(1,1)$

☐ $f(x,1)$

☒ $\frac{f(x,1)}{h(1)}$

☐ $\frac{f(x,1)}{h(x)}$

Made by: Waqar Siddhu

If $f(x, y)$ is bivariate probability density function of continuous random variables X and Y then marginal density function of y i.e. $h(y)$ is:

Answer (Please select your correct option)

VuAnswers.com

☐ $\int_{-\infty}^{\infty} f(x, y) dx$

☐ $\int_{-\infty}^{\infty} f(x, y) dy$

☒ $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy$

☐ $\int_a^b \int_c^d f(x, y) dy dx$

Made by: Waqar Siddhu

_____ is a range of numbers inferred from the sample that has a certain probability of including the population parameter over the long run.

Answer (Please select your correct option)

VuAnswers.com

☐ Hypothesis

☐ Lower limit

☒ Confidence interval

not sure

☐ Probability limit

Made by: Waqar Siddhu

How many parameter(s) are in t-distribution?

Answer (Please select your correct option)

VuAnswers.com

☐

0

☒

1

☐

2

☐

3

Made by: Waqar Siddhu

A judge can acquit a guilty person is the example of:

Answer (Please select your correct option)

VuAnswers.com

☐ Type I error

☐ Type II error

☒ Correct decision

not sure

☐ No information regarding this

Made by: Waqar Siddhu

If the sampling distribution of $\bar{X}$ is normal, we would expect 99% of the sample means to be within the interval:

Answer (Please select your correct option)

VuAnswers.com



$$\mu_x \pm 2\sigma_x$$



$$\mu_x \pm 1.96\sigma_x$$



$$\mu_x \pm 2.58\sigma_x$$



$$\mu_x \pm \sigma_x$$

Made by: Waqar Siddhu

The proportion of males in Pakistan is at least 0.48, the alternative hypothesis H_1 is

Answer (Please select your correct option)

VuAnswers.com

☐ $P \leq 0.48$

☐ $P = 0.48$

☒ $P < 0.48$

not sure

☐ $P \geq 0.48$

Made by: Waqar Siddhu

If mean of χ^2 distribution is k then variance will be:

Answer (Please select your correct option)

VuAnswers.com

☐

k^2

☒

$2k$

☐

$1/k$

☐

k

Made by: Waqar Siddhu

Test Statistics χ^2 is equal to:

Answer (Please select your correct option)

VuAnswers.com

☐ $\frac{n^2 S^2}{\sigma_0^2}$

☒ $\frac{n S^2}{\sigma_0^2}$

☐ $\frac{S^2}{2\sigma_0^2}$

☐ $\frac{n^2 S^2}{2\sigma_0^2}$

Made by: Waqar Siddhu

F- distribution is

Answer (Please select your correct option)

VuAnswers.com

☐ Positively skewed

☐ Negatively skewed

☐ Normal

☐ Symmetrical

Made by: Waqar Siddhu

The Chi- Square distribution is continuous distribution ranging from:

Answer (Please select your correct option)

VuAnswers.com

☐

$$-\infty \leq \chi^2 \leq \infty$$

☐

$$-\infty \leq \chi^2 \leq 1$$

☐

$$-\infty \leq \chi^2 \leq 0$$

☒

$$0 \leq \chi^2 \leq \infty$$

Made by: Waqar Siddhu

In Statistics, we have MSE which is abbreviation of

Answer (Please select your correct option)

☒ Mean square error

☐ Measured square error

☐ Medical screening exam

☐ Major sampling error

VuAnswers.com

Made by: Waqar Siddhu

The LSD test is applied when the null hypothesis is:

Answer (Please select your correct option)

☒ Rejected

☐ Accepted

☐ Finalized

☐ Acknowledged

VuAnswers.com

Made by: Waqar Siddhu

As the degree of freedom increases, the t-distribution tends to coincide with:

Answer (Please select your correct option)

VuAnswers.com

☐ Binomial distribution

☐ Uniform distribution

☐ Hypergeometric distribution

☒ Normal distribution

Made by: Waqar Siddhu

In a binomial experiment the total number of trials are:

Answer (Please select your correct option)

☒ Fixed in advance

☐ Changeable according to situation

☐ Unpredictable

☐ Not independent

VuAnswers.com

Made by: Waqar Siddhu

What is the probability of drawing a red-queen card from a well shuffled pack of 52 playing cards?

Answer (Please select your correct option)

VuAnswers.com

☐ 4/52

☒ 2/52

☐ 13/52

☐ 26/52

Made by: Waqar Siddhu

The probability of drawing a spade card is:

Answer (Please select your correct option)

VuAnswers.com

☐ 1/52

☐ 4/52

☒ 13/52

☐ 26/52

Made by: Waqar Siddhu

Which of the following is true for the binomial distribution $b(x, n, p)$:

Answer (Please select your correct option)

☒ mean > variance

☐ mean < variance

☐ mean = variance

☐ mean = standard deviation

VuAnswers.com

Made by: Waqar Siddhu

When $f(x)$ is continuous probability function, then $P(X = 2)$ is:

Answer (Please select your correct option)

VuAnswers.com

☐ 1

☐ 0.5

☒ 0

☐ 0.25

Made by: Waqar Siddhu

For any two estimators T_1 and T_2 , if $\text{VAR}(T_1) < \text{VAR}(T_2)$, then T_1 is:

Answer (Please select your correct option)

VuAnswers.com

☐ Unbaised

☐ Sufficient

☒ Efficient

☐ Consistent

Made by: Waqar Siddhu

In construction of a histogram, what would be taken along X-axis?

Answer (Please select your correct option)

VuAnswers.com

☐ Mid points

☐ Class limits

☐ Class interval

☒ Class boundaries

Made by: Waqar Siddhu

If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

Answer (Please select your correct option)

☐ Ogive

☐ Frequency polygon

☒ Frequency curve

☐ Histogram

VuAnswers.com

Made by: Waqar Siddhu

Tabulation is the process of arranging data into:

VuAnswers.com

Answer (Please select your correct option)

☐ Different classes

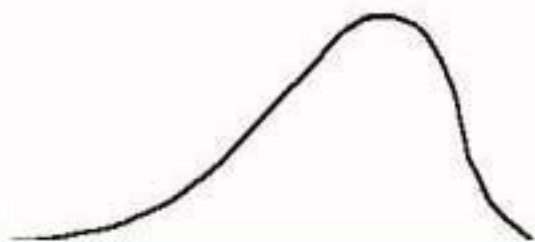
☐ Rows

☐ Columns

☒ Rows and Columns

Made by: Waqar Siddhu

Which type of the curve is represented by the following shape?



Answer (Please select your correct option)

☒ Negatively skewed curve

☐ Bell shape curve

☐ semi-symmetrical curve

☐ Positively skewed curve

VuAnswers.com

Made by: Waqar Siddhu

If $A = \{1, 2, 3, 4, 5, 10\}$ and $B = \{1, 3, 5\}$ then $B \subset A$ means:

Answer (Please select your correct option)

☐ A is less than B

☐ A is contained in B

☒ B is contained in A.

☐ B is less than A

VuAnswers.com

Made by: Waqar Siddhu

What is the mode in the word STATISTICS:

The Mode is defined as that value which occurs most frequently in a set of data i.e. it indicates the most common result.

Answer (Please select your correct option)

VuAnswers.com

☐ S

☐ T

☐ I

☒ S and T

Made by: Waqar Siddhu

In a symmetrical distribution, if $Q_1 = 20$, $Q_2 = 30$ then Q_3 is :

Answer (Please select your correct option)

VuAnswers.com

50



40



30



10



Made by: Waqar Siddhu

Which one of the following is not included in measures of central tendency:

Answer (Please select your correct option)

☒ Quartile Deviation

☐ Harmonic Mean

☐ Geometric Mean

☐ Arithmetic Mean

VuAnswers.com

Made by: Waqar Siddhu

The sum of deviations from mean is:

Answer (Please select your correct option)

☐ Maximum

☐ Minimum

☒ Zero

☐ Undefined

VuAnswers.com

Made by: Waqar Siddhu

Mean deviation is always:

Answer (Please select your correct option)

VuAnswers.com

☐ Less than Standard Deviation

☐ Greater than Standard Deviation

☐ Greater or equal to Standard Deviation

☒ Less or equal to Standard Deviation

Made by: Waqar Siddhu

The parameter of the chi- square distribution is.....

Answer (Please select your correct option)

VuAnswers.com

☒ ν

☐ $\nu - 1$

☐ $\nu - 2$

☐ $\nu - p$

Made by: Waqar Siddhu

The value of χ^2 can never be :

Answer (Please select your correct option)

VuAnswers.com

☐ Zero

☐ Less than 1

☐ Greater than 1

☒ Negative

Made by: Waqar Siddhu

The probability of an event always lies between:

Answer (Please select your correct option)

VuAnswers.com

☐ 0 and $-\infty$

☐ -1 and +1

☐ $-\infty$ and $+\infty$

☒ 0 and 1

Made by: Waqar Siddhu

The number of parameters in a Poisson distribution is (are):

Answer (Please select your correct option)

VuAnswers.com

☒ 1

☐ 0

☐ 2

☐ 3

Made by: Waqar Siddhu

$E(4X + 5) = \underline{\hspace{2cm}}$

Answer (Please select your correct option)

VuAnswers.com

☐ 16 E (X)☐ 16 E (X) + 5☐ 12 E (X)☒ 4 E (X) + 5

Made by: Waqar Siddhu

In a one-way ANOVA:

Answer (Please select your correct option)

VuAnswers.com

☐ The interaction term has $(c - 1)(n - 1)$ degrees of freedom

☐ An interaction term is given

☐ An interaction effect can be tested

☐ There is no interaction term

Made by: Waqar Siddhu

The degrees of freedom for a t-test with sample size 14 is:

Answer (Please select your correct option)

VuAnswers.com

☐ 14

☒ 13

☐ 7

☐ 0

Made by: Waqar Siddhu

The degrees of freedom for a t-test with sample size 6 is:

Answer (Please select your correct option)

VuAnswers.com

☐ 1

☐ 3

☒ 5

☐ 7

Made by: Waqar Siddhu



Which of the following is a characteristics of the normal distribution:

Answer (Please select your correct option)

VuAnswers.com

☐ It is a skewed distribution

☒ It is bell-shaped

☐ It is not asymptotic

☐ It is leptokurtic

Made by: Waqar Siddhu

In normal distribution $\beta_1 = \dots\dots\dots$

Answer (Please select your correct option)

VuAnswers.com

☒ 0

☐ 1

☐ 2

☐ 3

Made by: Waqar Siddhu

In normal distribution $\beta_2 = \dots\dots\dots$:

Answer (Please select your correct option)

VuAnswers.com

☐ 1

☐ 2

☒ 3

☐ 0

Made by: Waqar Siddhu

A good way to get a small standard error is to use a _____.

Answer (Please select your correct option)

VuAnswers.com

☐ Repeated sampling

☐ Small sample

☒ Large sample

☐ Large population

Made by: Waqar Siddhu

Which of the following is a measure of absolute dispersion?

Answer (Please select your correct option)

VuAnswers.com

☐

Skewness

☐

Mean Deviation

☐

Coefficient of variation

☐

Kurtosis

Made by: Waqar Siddhu

An automobile is running, during the first 60 Km, at the rate of 10 Km/hr. During the second 60 Km at the rate of 30Km/hr, while during the third 60 Km its speed was 40 Km/hr. What method is more appropriate to calculate the average speed?

Answer (Please select your correct option)

VuAnswers.com

☐ Median

☐ Arithmetic mean

☒ Harmonic mean

☐ Geometric mean

Made by: Waqar Siddhu

If $S.D(X) = 5$ then $S.D(\frac{2X+5}{2}) = \underline{\hspace{2cm}}$

Answer (Please select your correct option)

VuAnswers.com

☐ 5

☐ 10

☐ 15

☒ 7.5

Made by: Waqar Siddhu



For a particular data set the Pearson's coefficient of skewness is greater than zero. What will be the shape of distribution?

Answer (Please select your correct option)

VuAnswers.com

☐ Negatively skewed

☐ J-shaped

☐ Symmetrical

☒ Positively skewed

Made by: Waqar Siddhu

The total number of samples when sampling is done without replacement is equal to:

Answer (Please select your correct option)

VuAnswers.com

☒ N^n

☐ C_n^N

☐ $\frac{N-n}{N-1}$

☐ 1

Made by: Waqar Siddhu

Which of the following is correct option to find $P(X + Y \leq 1)$?

Answer (Please select your correct option)

VuAnswers.com

☐ $f(0, 0) + f(0, 1) + f(1, 2)$

☐ $f(2, 0) + f(0, 1) + f(1, 0)$

☐ $f(0, 0) + f(1, 1) + f(1, 0)$

☒ $f(0, 0) + f(0, 1) + f(1, 0)$

Made by: Waqar Siddhu

The conditional probability function $f(x|1) =$ _____

Answer (Please select your correct option)

VuAnswers.com

☐ $f(1,1)$

☐ $f(x,1)$

☒ $\frac{f(x,1)}{h(1)}$

☐ $\frac{f(x,1)}{h(x)}$

Made by: Waqar Siddhu

If $f(x, y)$ is bivariate probability density function of continuous random variables X and Y then marginal density function of y i.e. $h(y)$ is:

Answer (Please select your correct option)

VuAnswers.com

☐ $\int_{-\infty}^{\infty} f(x, y) dx$

☐ $\int_{-\infty}^{\infty} f(x, y) dy$

☒ $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy$

☐ $\int_a^b \int_c^d f(x, y) dy dx$

Made by: Waqar Siddhu



_____ is a range of numbers inferred from the sample that has a certain probability of including the population parameter over the long run.

Answer (Please select your correct option)

VuAnswers.com

- ☐ Hypothesis
- ☐ Lower limit
- ☒ Confidence interval
- ☐ Probability limit

not sure

Made by: Waqar Siddhu

By definition $f(y | x) =$ _____

Answer (Please select your correct option)

VuAnswers.com

☐ $f(y)$

☐ $f(x, y)$

☒ $\frac{f(x, y)}{h(x)}$

☐ $\frac{f(x, y)}{h(y)}$

Made by: Waqar Siddhu

Which of the following is an unbiased estimator of population σ^2 ?

Answer (Please select your correct option)

VuAnswers.com

☐ $\frac{\sum (x - \bar{x})^2}{n}$

☐ $\frac{\sum (x - \bar{x})^2}{n - 2}$

☐ $\frac{\sum (x - \bar{x})^2}{n(n - 1)}$

☒ $\frac{\sum (x - \bar{x})^2}{n - 1}$

Made by: Waqar Siddhu

If a significance level of 1% is used rather than 5%, the null hypothesis is:

Answer (Please select your correct option)

VuAnswers.com

☐ More likely to be rejected

☒ Less likely to be rejected

☐ Just as likely to be rejected

☐ None of the above

Made by: Waqar Siddhu

For $\alpha = 0.01$, the critical values of z for two tailed test are equal to:

Answer (Please select your correct option)

VuAnswers.com

☒ -2.58 and +2.58

☐ -2.33 and +2.33

☐ -1.645 and +1.645

☐ -1.96 and + 1.96

Made by: Waqar Siddhu

A judge can acquit a guilty person is the example of:

Answer (Please select your correct option)

VuAnswers.com

☐ Type I error

☐ Type II error

☒ Correct decision

idea

☐ No information regarding this

Made by: Waqar Siddhu

The proportion of males in Pakistan is at least 0.48, the alternative hypothesis H_1 is

Answer (Please select your correct option)

VuAnswers.com

☐ $P \leq 0.48$

☐ $P = 0.48$

☒ $P < 0.48$

idea

☐ $P \geq 0.48$

Made by: Waqar Siddhu

If mean of χ^2 distribution is k then variance will be:

Answer (Please select your correct option)

VuAnswers.com

☐ k^2

☒ $2k$

☐ $1/k$

☐ k

Made by: Waqar Siddhu

What is the graphical shape of the chi-square distribution?

Answer (Please select your correct option)

VuAnswers.com

☒ Positively skewed

☐ Negatively skewed

☐ Uniformly distributed

☐ Normally distributed

Made by: Waqar Siddhu

What factor determines the shape of the t-distribution?

Answer (Please select your correct option)

VuAnswers.com

☒ Degree of freedom

☐ Critical value

☐ Frequency of data

☐ Probability

Made by: Waqar Siddhu



The covariance of a random variable with itself is:

Answer (Please select your correct option)

VuAnswers.com

Zero

☐

One

☐

Its variance

☒

Its correlation

☐

Made by: Waqar Siddhu

Hypergeometric probability distribution has :

Answer (Please select your correct option)

VuAnswers.com

☐ (n, k) parameter

☐ (N) parameter

☐ (N, n, N-k) parameter

☒ (N, n, k) parameter

Made by: Waqar Siddhu

For any two estimators T_1 and T_2 , if $\text{VAR}(T_1) < \text{VAR}(T_2)$, then T_1 is:

Answer (Please select your correct option)

VuAnswers.com

☐ Consistent

☐ Unbiased

☐ Sufficient

☒ Efficient

Made by: Waqar Siddhu

Atmosphere pressure is the example of:

Answer (Please select your correct option)

VuAnswers.com

☐ Constant

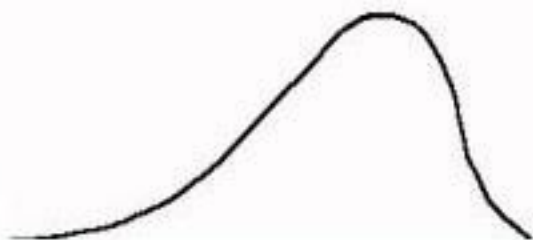
☐ Qualitative variable

☒ Quantitative variable

☐ None of the above

Made by: Waqar Siddhu

Which type of the curve is represented by the following shape?



Answer (Please select your correct option)

VuAnswers.com

☐ Bell shape curve

☐ semi-symmetrical curve

☐ Positively skewed curve

☒ Negatively skewed curve

Made by: Waqar Siddhu

Which one of the following is not a type of frequency curve?

Answer (Please select your correct option)

VuAnswers.com

- ☐ The symmetrical frequency curve
- ☐ The extremely skewed frequency curve
- ☐ The U-shaped frequency curve
- ☒ Frequency polygon

Made by: Waqar Siddhu

For positively skewed distribution
Mean.....Median.....Mode:

Answer (Please select your correct option)

VuAnswers.com

☐

μ

☐

Π

☐

$\wedge$

☒

$\vee$

Made by: Waqar Siddhu

What is ' f_m ' in the formula of mode:

Answer (Please select your correct option)

VuAnswers.com

☐ First Frequency

☐ Last Frequency

☒ Maximum Frequency

☐ Minimum Frequency

Made by: Waqar Siddhu

Which one is the measure of central tendency:

Answer (Please select your correct option)

VuAnswers.com

☐ Variation of the distribution

☒ Average of the distribution

☐ Scatterness of the distribution

☐ Dispersion of the distribution

Made by: Waqar Siddhu

In a set of 10 values all the values are 5, what will be the P_{50} ?

Answer (Please select your correct option)

VuAnswers.com

2

☐

5

☒

i think

10

☐

20

☐

Made by: Waqar Siddhu

The parameter of the chi- square distribution is.....

Answer (Please select your correct option)

VuAnswers.com

☒ ν

☐ $\nu - 1$

☐ $\nu - 2$

☐ $\nu - p$

Made by: Waqar Siddhu

The value of χ^2 can never be :

Answer (Please select your correct option)

VuAnswers.com

☐ Zero

☐ Less than 1

☐ Greater than 1

☒ Negative

Made by: Waqar Siddhu

Analysis of variance is a procedure that enables us to test the equality of several:

Answer (Please select your correct option)

VuAnswers.com

☐ Variances

ANALYSIS OF VARIANCE (ANOVA)

It is a procedure which enables us to test the hypothesis of equality of several population means

(i.e.

$H_0 : \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$

against

☒ Means

☐ Proportions

☐ Groups

Made by: Waqar Siddhu

When two coins are tossed the probability of at most one head is:

Answer (Please select your correct option)

VuAnswers.com

☐ 1/4

☒ 2/4

☐ 3/4

☐ 4/4

Made by: Waqar Siddhu

The critical region for $H_1: \mu < \mu_0$ when $\alpha = 0.05$ is:

Answer (Please select your correct option)

VuAnswers.com

☒ $z > z_{0.05}$

☐ $|z| > z_{0.10}$

☐ $z < -z_{0.05}$

☐ $|z| > z_{0.05}$

Made by: Waqar Siddhu

A discrete probability function $f(x)$ is always:

Answer (Please select your correct option)

VuAnswers.com

☐ Non-negative

☐ Negative

☒ One

☐ Zero

Made by: Waqar Siddhu

When $f(x)$ is continuous probability function, then $P(X = 1)$ is:

Answer (Please select your correct option)

VuAnswers.com

☐

1

☐

∞

☐

$-\infty$

☒

0

Made by: Waqar Siddhu

A random variable can be generated:

Answer (Please select your correct option)

VuAnswers.com

☐ Manually

☐ Mechanically

Not sure.. :)

concept.. :)

☒ Manually & Mechanically

☐ Mathematically

numerical quantity whose value is determined by the outcome of a random experiment is called a random variable.

Made by: Waqar Siddhu

The curve of the F- distribution depends upon:

Answer (Please select your correct option)

VuAnswers.com

☒ Degrees of freedom

☐ Standard deviation

☐ Mean

☐ Variance

Made by: Waqar Siddhu

Assuming that following is a probability distribution, then what is the value of 'a':

X	1	2	3
P(X)	0.1	a	0.1

Answer (Please select your correct option)

VuAnswers.com

☐ 0.6

☒ 0.8

☐ 0.2

☐ 0.4

Made by: Waqar Siddhu

The mode value from raw data can be obtained by the help of

Answer (Please select your correct option)

VuAnswers.com

☐ Dot plot

☐ Stem and leaf plot

☐ Bar chart

☒ None of these

Made by: Waqar Siddhu

In normal distribution, the quartile deviation Q.D =

Answer (Please select your correct option)

VuAnswers.com

- ☐ 0.5σ
- ☐ 0.75σ
- ☐ 0.7979σ
- ☒ 0.6745σ

Made by: Waqar Siddhu

A randomly selected sample of 400 students at university was asked whether or not they will participate in politics. Forty-six percent of the 400 student surveyed answered "yes". Which one of the following statement about number 46% is correct?

Answer (Please select your correct option)

VuAnswers.com

☐ It is a sample statistic.

☐ It is a population parameter.

☐ It is a margin of error.

☒ It is a standard error.

not sure

Made by: Waqar Siddhu

A student solved 25 questions from first 50 questions of a book . The probability that he will solve the remaining all questions is:

Answer (Please select your correct option)

VuAnswers.com

☐ 0.25

☒ 0.5

☐ 1

☐ 0

Made by: Waqar Siddhu

The average which is defined as the reciprocal of the arithmetic mean of the reciprocals of the values is called:

Answer (Please select your correct option)

VuAnswers.com

☐ Geometric Mean

☒ Harmonic Mean

☐ Mode

☐ Median

Made by: Waqar Siddhu

For the independent events A and B if $P(A) = 0.25$, $P(B) = 0.40$ then $P(A \cap B) = \dots\dots\dots$

Answer (Please select your correct option)

VuAnswers.com

☐ 0.65

☒ 0.1

☐ 0.50

☐ 0.15

Made by: Waqar Siddhu

The total number of samples when sampling is done with replacement is equal to:

Answer (Please select your correct option)

VuAnswers.com

☒ N^n

☐ C_n^N

☐ $\frac{N-n}{N-1}$

☐ 1

Made by: Waqar Siddhu

Which of the following is correct option to find $P(X + Y \leq 1)$?

Answer (Please select your correct option)

VuAnswers.com

☐ $f(0, 0) + f(0, 1) + f(1, 2)$

☐ $f(2, 0) + f(0, 1) + f(1, 0)$

☐ $f(0, 0) + f(1, 1) + f(1, 0)$

☒ $f(0, 0) + f(0, 1) + f(1, 0)$

Made by: Waqar Siddhu

The conditional probability function $f(x|1) =$ _____

Answer (Please select your correct option)

VuAnswers.com

☐ $f(1,1)$

☐ $f(x,1)$

☒ $\frac{f(x,1)}{h(1)}$

☐ $\frac{f(x,1)}{h(x)}$

Made by: Waqar Siddhu

In interval estimation, we always get:

Answer (Please select your correct option)

VuAnswers.com

☒ Range of values

☐ Zero

☐ Single value

☐ Two values

Made by: Waqar Siddhu

Which of the following is an unbiased estimator of population σ^2 ?

VuAnswers.com

Answer (Please select your correct option)

☐ $\frac{\sum (x - \bar{x})^2}{n}$

☐ $\frac{\sum (x - \bar{x})^2}{n - 2}$

☐ $\frac{\sum (x - \bar{x})^2}{n(n - 1)}$

☒ $\frac{\sum (x - \bar{x})^2}{n - 1}$

Made by: Waqar Siddhu

How many parameter(s) are in t-distribution?

Answer (Please select your correct option)

VuAnswers.com

0

☐

1

☒

2

☐

3

☐

Made by: Waqar Siddhu

A deserving player is not selected in the team is an example of:

Answer (Please select your correct option)

VuAnswers.com

☐ Type I error

☒ Type II error

☐ Correct decision

☐ No information regarding this

Made by: Waqar Siddhu

After an interval is constructed, then what is the probability of occurrence of the parameter in it?

Answer (Please select your correct option)

VuAnswers.com

☐ Zero

☐ One

☒ Fifty

☐ Either one or zero

Made by: Waqar Siddhu

If $\bar{X}$ is the mean of the n observations, then which test statistic will be used to calculate the confidence limits of the population variance σ^2 ?

Answer (Please select your correct option)

VuAnswers.com

☐ Z-statistic

☐ T-statistic

☒ χ^2 -statistics

☐ F-statistics

Made by: Waqar Siddhu

If mean of χ^2 distribution is k then variance will be:

Answer (Please select your correct option)

VuAnswers.com

☐ k^2 ☒ $2k$ ☐ $1/k$ ☐ k

Made by: Waqar Siddhu

What is the graphical shape of the chi-square distribution?

Answer (Please select your correct option)

VuAnswers.com

☒ Positively skewed

☐ Negatively skewed

☐ Uniformly distributed

☐ Normally distributed

Made by: Waqar Siddhu

What factor determines the shape of the t-distribution?

Answer (Please select your correct option)

VuAnswers.com

☒ Degree of freedom

☐ Critical value

☐ Frequency of data

☐ Probability

Made by: Waqar Siddhu

If X and Y are random variables, then $E(X - Y)$ is equal to:

Answer (Please select your correct option)

VuAnswers.com

☐ $E(X) + E(Y)$

☒ $E(X) - E(Y)$

☐ $X - E(Y)$

☐ $E(X) - Y$

Made by: Waqar Siddhu

In the regression line $Y = a + bX$, the non-random variable is:

Answer (Please select your correct option)

VuAnswers.com

not sure

☒ Y

☐ X

☐ Both X and Y

☐ Neither X nor Y

Made by: Waqar Siddhu

The parameters of the binomial distribution $b(x, n, p)$ are:

Answer (Please select your correct option)

VuAnswers.com

☐ x & n

☐ x & p

☒ n & p

☐ x, n & p

Made by: Waqar Siddhu

Which of the following is true for the binomial distribution $b(x, n, p)$:

Answer (Please select your correct option)

VuAnswers.com

☒ mean > variance

☐ mean < variance

☐ mean = variance

☐ mean = standard deviation

Made by: Waqar Siddhu

Hypergeometric probability distribution has :

Answer (Please select your correct option)

VuAnswers.com

- ☐ (n, k) parameter
- ☐ (N) parameter
- ☐ (N, n, N-k) parameter
- ☒ (N, n, k) parameter

Made by: Waqar Siddhu

The variance of the hypergeometric probability distribution is:

VuAnswers.com

Answer (Please select your correct option)

☐ $\sigma^2 = n \frac{k}{N}$

☒ $\sigma^2 = n \frac{k}{N} \frac{N-k}{N} \frac{N-n}{N-1},$

☐ $\sigma^2 = npq$

☐ $\sigma^2 = N \frac{n}{K}$

Made by: Waqar Siddhu

Tabulation is the process of arranging data into:

Answer (Please select your correct option)

VuAnswers.com

☐ Different classes

☐ Rows

☐ Columns

☒ Rows and Columns

Made by: Waqar Siddhu

A simple bar chart consists of:

Answer (Please select your correct option)

VuAnswers.com

☐ Vertical bars

☐ Horizontal bars

☒ Vertical or horizontal bars

☐ Multiple bars

Made by: Waqar Siddhu

Which one is the formula to calculate the approximate value of class interval?

Answer (Please select your correct option)

VuAnswers.com

☐ Minimum value/Range

☐ Maximum value/No. of classes

☐ No. of classes/Range

☒ Range/No. of classes

Made by: Waqar Siddhu

Which one of the following is equal to the 2nd quartile:

Answer (Please select your correct option)

VuAnswers.com



P_{33}



D_3



Median



Mode

Made by: Waqar Siddhu

For the independent events A and B if $P(A) = 0.15$, $P(B) = 0.50$ then $P(A \cap B) = \dots\dots\dots$

Answer (Please select your correct option)

VuAnswers.com

☐ 0.50

☒ 0.075

☐ 0.125

☐ 0.060

Made by: Waqar Siddhu

Which one of the following measure is not based on all the observations?

Answer (Please select your correct option)

VuAnswers.com

☐ Arithmetic Mean

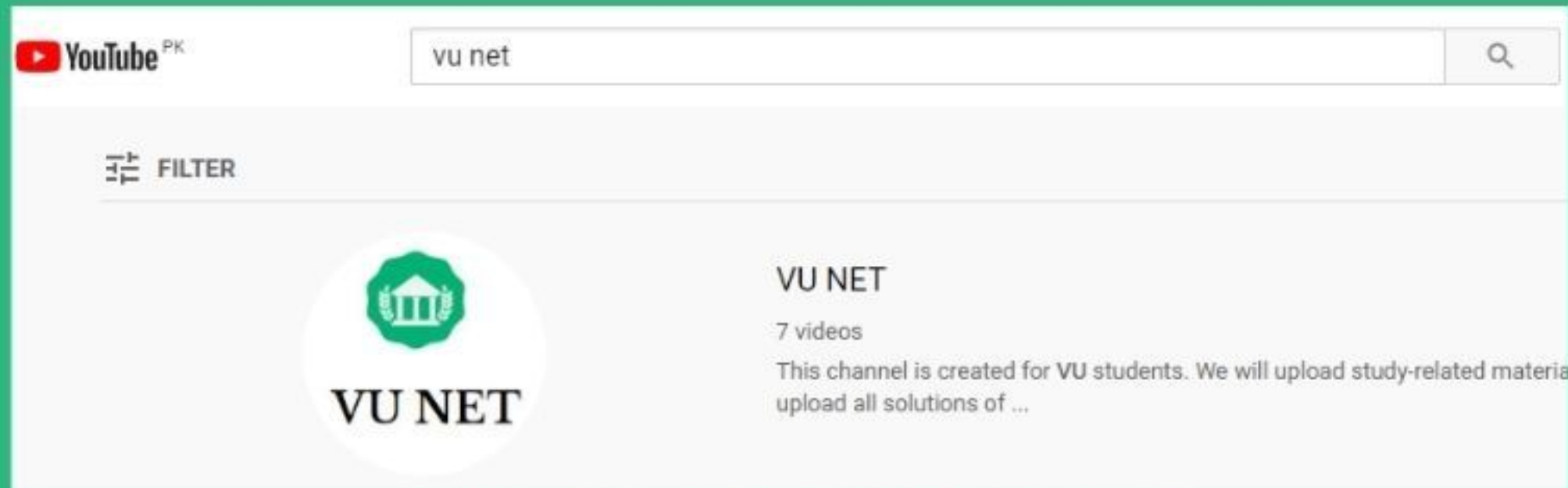
☐ Geometric Mean

☐ Harmonic Mean

☒ Mode

Made by: Waqar Siddhu

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